

Aman Sharma

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I'm

Robotics enthusiast focused on 3D modeling, mechanical design, and hardware prototyping. Hands-on experience designing and building robotic systems, mechanical structures, and RC models. Interested in transforming ideas into functional physical prototypes through CAD, fabrication, electronics, and iterative testing.

Education

February 2025

Hydrabad, India

Birla Institute of Technology, Hyderabad

BSc(Hons) in Computer Science

- **Relevant Coursework:** Equivalency for any prerequisites for a 4-year undergraduate programme. Advance Knowledge in Computer Application & Machine Learning.

Skills

3D Design & Mechanical

- 3D Modeling & CAD
- Mechanical Design Fundamentals
- Product & Part Modeling
- Mechanical Assemblies
- Basic Design for Fabrication
- Rapid Prototyping

Robotics & Hardware

- Robotics Fundamentals
- Hardware Prototyping
- Embedded Systems
- Sensor Integration
- Motors & Actuators
- Basic Automation & Control

Software & Programming

- Fusion 360
- Arduino / ESP32
- C, C++, Python
- HTML, CSS, JavaScript
- MS Office
- Canva / Inkscape

Projects | 3D Design & Robotics

[Portfolio](#)

Mobile Robotic Manipulator – 6WD UGV & Robotic Arm

- Built a 6WD UGV with an integrated 6-DOF robotic arm for mobile manipulation.
- 3D-modeled the robotic arm base/mount, focusing on support, alignment, and chassis integration.
- Integrated ESP32 wireless control, motors, servos, sensors, and safety systems.

Quadruped Robot

- Designed and built a quadruped robot mechanical structure, focusing on 3D modeling, leg geometry, stability, and load distribution.
- Developed a low-cost mechanical chassis and iterated the structure for improved walking stability and future sensor integration.

F-22 Raptor RC Aircraft

- Designed and built a lightweight F-22 Raptor model using thermocol, focusing on structure and balance.
- Implemented wireless control using FlySky RC transmitter with FS-iA10B receiver for aircraft maneuvering.

Hand Gesture Controlled Robotics Hand

- Built a gesture-controlled robotic hand using OpenCV, Arduino, and servo motors for real-time finger motion control.
- Designed and fabricated the robotic hand using cardboard/plastic, mechanical finger joints, and flexible thread-based linkages.

Handmade Carbon Fiber Drone

Self-Balancing Robot

Line Following Robot

and more...

Experience

Mechanical Design Intern (3 Months, On-site)

Beckkon Systems Pvt. Ltd. (IIT BHU, Varansi)

Develop 3D CAD models and mechanical designs for hardware applications. Refine designs through iterations, feedback, and prototyping.

Graphic Designer (2 Months, On-site)

Marketing Intern as Graphic Designer (Bengaluru)

Designed posters, social media creatives, and digital assets using Canva, Inkscape, and MS Office tools. Collaborated with the team to develop consistent and visually effective content.

Neura robotics

STEM - Student Exploring Robotics Innovation

I am a robotics enthusiast with a basic understanding of programming and electronics. I enjoy learning how robots work and have participated in small team projects using Arduino and sensors. I'm excited to grow my skills and contribute to real-world robotics solutions.

HTML, CSS

Self-Learning

Skilled in creating simple and responsive web pages using HTML, CSS for adding interactivity. Creative mindset with a passion for designing and building things differently. Some basic Knowledge of **BOOTSTRAP**.